

Evidencia de ejecución

Laboratorio PCAM + OpenMP · Simulación N-Body 2D · Nicolás Concua · Diego · Esteban Cárcamo

Todas las capturas son de la terminal del mismo equipo en el que se generó `results.csv`. La salida completa de cada captura está en `evidence/*.txt`, y la de cada corrida de los experimentos, en `results/raw/`. Parámetros constantes: **N = 20 000, 10 pasos, semilla 42**.

Entorno

```
date: 2026-10-04T22:50:44-06:00
N: 20000 | Steps: 10 | Seed: 42 | Runs: 3
host: Linux 7.2.6-1-cachyos x86_64
CPU(s): 32
Model name: 13th Gen Intel(R) Core(TM) i9-13980HX
Thread(s) per core: 2
Core(s) per socket: 24
Socket(s): 1
compiler: cc (GCC) 16.2.1 20260810
Processors reported by OpenMP: 32
Max threads reported by OpenMP: 32
```

Evidencia 1 · Compilación exitosa

```
Evidencia 1 - Compilación

escu@escu:~/Laboratorio_PCAM$ make clean && make
rm -rf bin
mkdir -p bin
cc -O2 -std=c11 -Wall -Wextra -DSTUDENT="\Nicolas Concua, Diego, Esteban\" -fopenmp src/compare_states.c -o
  bin/compare_states -fopenmp -lm
cc -O2 -std=c11 -Wall -Wextra -DSTUDENT="\Nicolas Concua, Diego, Esteban\" -fopenmp src/nbody_paralelo.c -o
  bin/nbody_paralelo -fopenmp -lm
cc -O2 -std=c11 -Wall -Wextra -DSTUDENT="\Nicolas Concua, Diego, Esteban\" -fopenmp src/nbody_secuencial.c
  -o bin/nbody_secuencial -fopenmp -lm
escu@escu:~/Laboratorio_PCAM$ ls -l bin/
total 64
-rwxr-xr-x 1 escu escu 16648 oct  4 22:57 compare_states
-rwxr-xr-x 1 escu escu 21912 oct  4 22:57 nbody_paralelo
-rwxr-xr-x 1 escu escu 16776 oct  4 22:57 nbody_secuencial
```

`make clean && make` compila las tres herramientas con `-Wall -Wextra` sin errores ni advertencias.

Evidencia 2 · Ejecución secuencial (A)

```
Evidencia 2 - Ejecución secuencial (A)

escu@escu:~/Laboratorio_PCAM$ ./bin/nbody_secuencial 20000 10
Student: Nicolas Concua, Diego, Esteban
Executable: ./bin/nbody_secuencial
Mode: sequential
N: 20000 | Steps: 10 | Seed: 42
Threads requested: 1 | Threads used: 1
Schedule: none | Chunk: -
Elapsed: 9.383 s
Checksum: 21071.127819
Processors reported by OpenMP: 32
Max threads reported by OpenMP: 32
```

N, pasos, tiempo y checksum de la versión base.

Evidencia 3 · Ejecución paralela directa (B)

```
Evidencia 3 - Ejecución paralela directa (B)

escu@escu:~/Laboratorio_PCAM$ ./bin/nbody_paralelo 20000 10 --mode parallel --threads 8 --schedule dynamic
--chunk 64
Student: Nicolas Concua, Diego, Esteban
Executable: ./bin/nbody_paralelo
Mode: parallel
N: 20000 | Steps: 10 | Seed: 42
Threads requested: 8 | Threads used: 8
Schedule: dynamic | Chunk: 64
Elapsed: 1.628 s
Checksum: 21071.127819
Processors reported by OpenMP: 32
Max threads reported by OpenMP: 32
```

Threads pedidos y usados, schedule, chunk, tiempo y checksum.

Evidencia 4 · Ejecución optimizada por pares (C)

```
Evidencia 4 - Ejecución optimizada por pares (C)

escu@escu:~/Laboratorio_PCAM$ ./bin/nbody_paralelo 20000 10 --mode optimized --threads 8 --schedule dynamic
--chunk 64
Student: Nicolas Concua, Diego, Esteban
Executable: ./bin/nbody_paralelo
Mode: parallel-optimized
N: 20000 | Steps: 10 | Seed: 42
Threads requested: 8 | Threads used: 8
Schedule: dynamic | Chunk: 64
Elapsed: 0.888 s
Checksum: 21071.127819
Processors reported by OpenMP: 32
Max threads reported by OpenMP: 32
```

Simetría de pares con acumuladores privados por thread; mismo checksum que la secuencial.

Evidencia 5 · Resumen de hardware

```
Evidencia 5 - Resumen de hardware

escu@escu:~/Laboratorio_PCAM$ LC_ALL=C lscpu | grep -E '^(Model name|CPU(s)|Thread(s) per core|Core(s) per socket):'
CPU(s): 32
Model name: 13th Gen Intel(R) Core(TM) i9-13980HX
Thread(s) per core: 2
Core(s) per socket: 24
escu@escu:~/Laboratorio_PCAM$ ./bin/nbody_paralelo 20000 1 --mode parallel --threads 8 | grep -E 'Threads|Processors|Max threads'
Threads requested: 8 | Threads used: 8
Processors reported by OpenMP: 32
Max threads reported by OpenMP: 32
```

`omp_get_num_procs()` y `omp_get_max_threads()` impresos por el programa, junto con `lscpu`.

Corrección contra la secuencial

```
Extra - Corrección contra la secuencial

escu@escu:~/Laboratorio_PCAM$ ./scripts/check_correctness.sh 20000 10 8
= Secuencial (referencia) =
Mode: sequential
Checksum: 21071.127819

= parallel (8 threads) =
Mode: parallel
Checksum: 21071.127819
Bodies compared: 20000
Max abs diff: 0.000e+00 (body 0, x)
Tolerance: 1.0e-09 → OK

= optimized (8 threads) =
Mode: parallel-optimized
Checksum: 21071.127819
Bodies compared: 20000
Max abs diff: 1.197e-16 (body 5057, vy)
Tolerance: 1.0e-09 → OK
```

Diferencia máxima del estado final (x, y, vx, vy): 0 en B y $1,2 \cdot 10^{-16}$ en C, por el cambio de orden en las sumas de punto flotante.

Tiempos promedio, speedup y efficiency

Promedio de 3 corridas por configuración. $S = T_1 / T_p$ con T_1 = tiempo de la secuencial; $E = S / p$.

Threads	Schedule	Chunk	$\bar{T}_B$ (s)	$\bar{T}_C$ (s)	S / E_B	S / E_C
1	sequential	-	9.362	—	1.00 / 1.00	—
2	static	default	4.966	4.097	1.89 / 0.94	2.29 / 1.14
4	static	default	3.038	2.464	3.08 / 0.77	3.80 / 0.95
8	static	default	1.723	1.475	5.43 / 0.68	6.35 / 0.79
8	static	8	1.789	1.061	5.23 / 0.65	8.82 / 1.10
8	static	64	1.800	1.089	5.20 / 0.65	8.60 / 1.07
8	dynamic	8	1.675	0.962	5.59 / 0.70	9.73 / 1.22
8	dynamic	64	1.680	0.965	5.57 / 0.70	9.70 / 1.21
8	guided	default	1.714	1.485	5.46 / 0.68	6.30 / 0.79

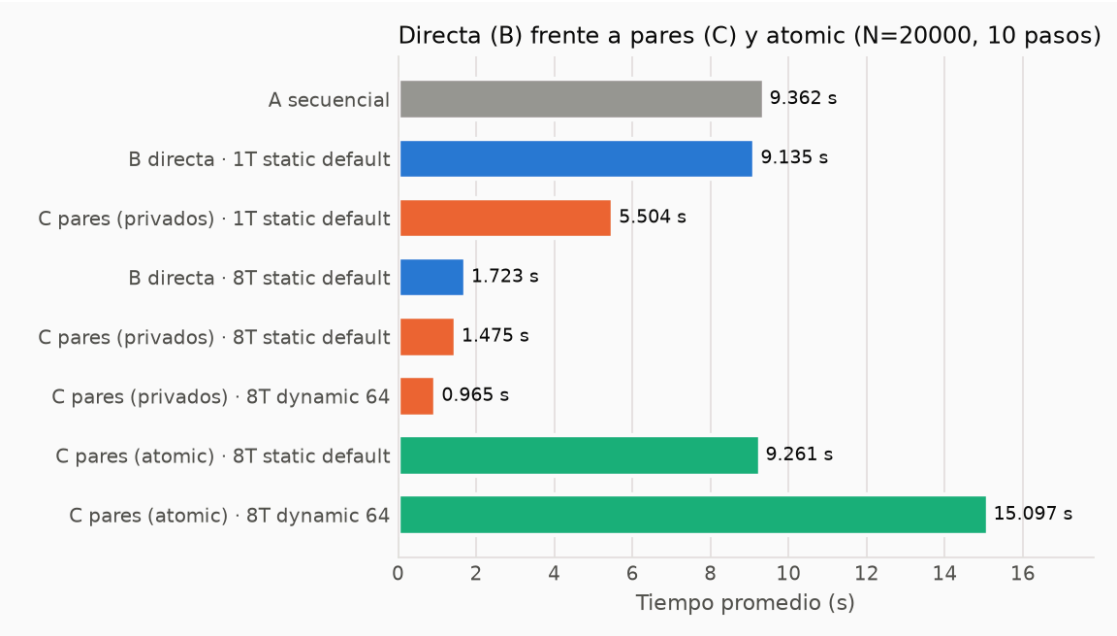
Schedule y chunk con 8 threads

Schedule	Chunk	$\bar{T}_B$ (s)	S / E_B	$\bar{T}_C$ (s)	S / E_C
static	default	1.723	5.43 / 0.68	1.475	6.35 / 0.79
static	8	1.789	5.23 / 0.65	1.061	8.82 / 1.10
static	64	1.800	5.20 / 0.65	1.089	8.60 / 1.07
dynamic	default	1.667	5.62 / 0.70	0.958	9.77 / 1.22
dynamic	8	1.675	5.59 / 0.70	0.962	9.73 / 1.22
dynamic	64	1.680	5.57 / 0.70	0.965	9.70 / 1.21
guided	default	1.714	5.46 / 0.68	1.485	6.30 / 0.79
guided	8	1.694	5.53 / 0.69	1.518	6.17 / 0.77
guided	64	1.727	5.42 / 0.68	1.507	6.21 / 0.78

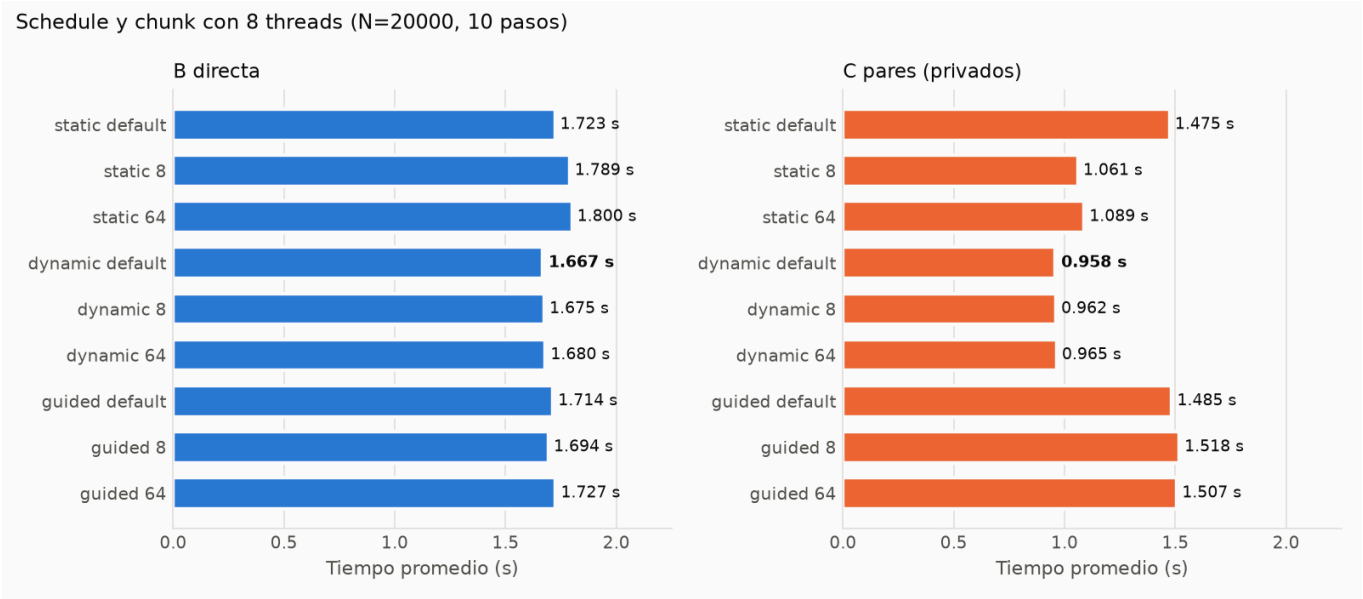
Escalabilidad (static default)

Threads	$\bar{T}_B$ (s)	S / E_B	$\bar{T}_C$ (s)	S / E_C
1	9.135	1.02 / 1.02	5.504	1.70 / 1.70
2	4.966	1.89 / 0.94	4.097	2.29 / 1.14
4	3.038	3.08 / 0.77	2.464	3.80 / 0.95
8	1.723	5.43 / 0.68	1.475	6.35 / 0.79
16	1.477	6.34 / 0.40	1.061	8.82 / 0.55
32	1.127	8.31 / 0.26	0.823	11.38 / 0.35

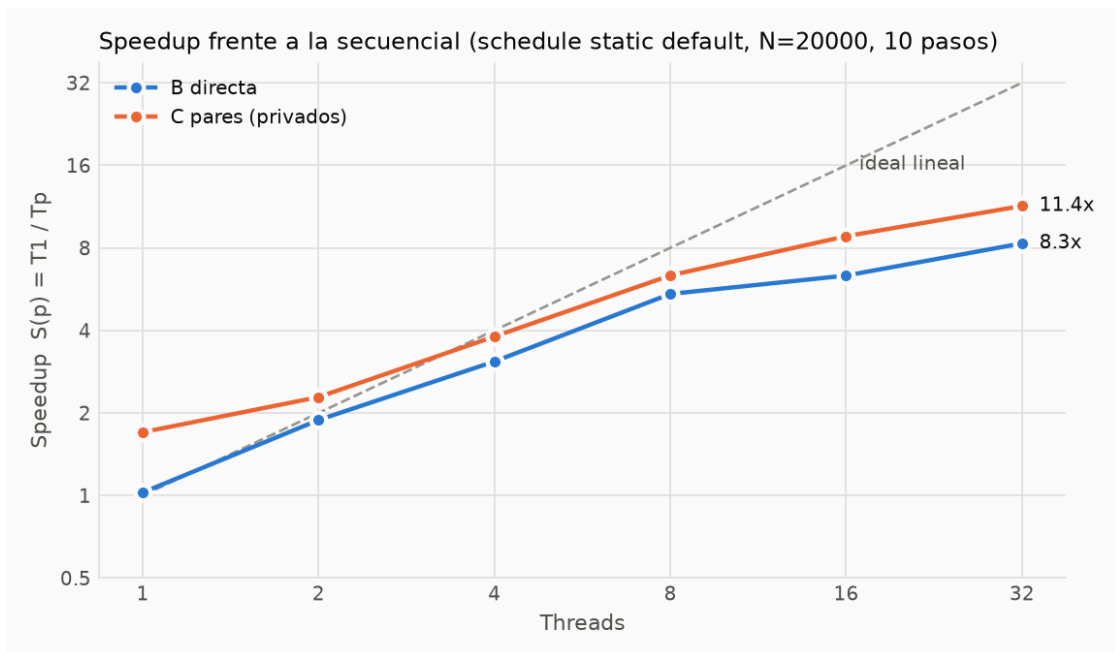
Comparaciones de rendimiento



Directa (B) frente a pares (C): con 8 threads y dynamic, C es 1,74× más rápida que la mejor B; la variante atomic es más lenta que la secuencial.



En B el schedule casi no importa (carga uniforme); en C, static default y guided sufren el desbalance triangular.



Speedup frente a threads. A partir de 8 se agotan los núcleos de rendimiento del i9-13980HX.

Anexo · results.csv completo

Versión	T	Schedule	Chunk	Run 1	Run 2	Run 3	$\bar{T}$ (s)	S	E	Checksum
A secuencial	1	sequential	-	9.390	9.416	9.281	9.362	1.00	1.00	21071.127819
B directa	1	static	default	9.584	8.936	8.885	9.135	1.02	1.02	21071.127819
B directa	2	static	default	4.899	5.025	4.975	4.966	1.89	0.94	21071.127819
B directa	4	static	default	3.033	3.097	2.985	3.038	3.08	0.77	21071.127819
B directa	8	static	default	1.686	1.741	1.743	1.723	5.43	0.68	21071.127819
B directa	8	static	8	1.770	1.786	1.812	1.789	5.23	0.65	21071.127819
B directa	8	static	64	1.806	1.763	1.832	1.800	5.20	0.65	21071.127819
B directa	8	dynamic	default	1.663	1.672	1.666	1.667	5.62	0.70	21071.127819
B directa	8	dynamic	8	1.677	1.662	1.687	1.675	5.59	0.70	21071.127819
B directa	8	dynamic	64	1.685	1.676	1.680	1.680	5.57	0.70	21071.127819
B directa	8	guided	default	1.676	1.737	1.730	1.714	5.46	0.68	21071.127819
B directa	8	guided	8	1.707	1.701	1.675	1.694	5.53	0.69	21071.127819
B directa	8	guided	64	1.713	1.740	1.727	1.727	5.42	0.68	21071.127819
B directa	16	static	default	1.472	1.480	1.478	1.477	6.34	0.40	21071.127819
B directa	32	static	default	1.125	1.127	1.128	1.127	8.31	0.26	21071.127819
C pares	1	static	default	5.683	5.595	5.234	5.504	1.70	1.70	21071.127819
C pares	2	static	default	4.192	4.303	3.796	4.097	2.29	1.14	21071.127819
C pares	4	static	default	2.444	2.446	2.503	2.464	3.80	0.95	21071.127819
C pares	8	static	default	1.445	1.439	1.541	1.475	6.35	0.79	21071.127819
C pares	8	static	8	1.053	1.086	1.044	1.061	8.82	1.10	21071.127819
C pares	8	static	64	1.102	1.070	1.095	1.089	8.60	1.07	21071.127819
C pares	8	dynamic	default	0.931	0.968	0.976	0.958	9.77	1.22	21071.127819
C pares	8	dynamic	8	0.953	0.974	0.958	0.962	9.73	1.22	21071.127819
C pares	8	dynamic	64	0.960	0.969	0.966	0.965	9.70	1.21	21071.127819
C pares	8	guided	default	1.484	1.451	1.520	1.485	6.30	0.79	21071.127819
C pares	8	guided	8	1.523	1.497	1.535	1.518	6.17	0.77	21071.127819
C pares	8	guided	64	1.503	1.558	1.461	1.507	6.21	0.78	21071.127819
C pares	16	static	default	1.068	1.072	1.043	1.061	8.82	0.55	21071.127819
C pares	32	static	default	0.796	0.875	0.798	0.823	11.38	0.35	21071.127819
C atomic	8	static	default	9.626	9.276	8.881	9.261	1.01	0.13	21071.127819
C atomic	8	dynamic	64	14.444	15.041	15.805	15.097	0.62	0.08	21071.127819